

From Web Collection to Corpus Analysis: Patriotic Alternative News

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Code and outputs: github.com/Nouran-Khallaf/pa-news-multimodal-corpus

Proposed GitHub Pages dashboard: nouran-khallaf.github.io/pa-news-multimodal-corpus/

1. Executive summary

I chose Patriotic Alternative from the four candidate websites because its public, paginated news archive offered the richest combination of article text and images. The finished corpus holds 1,801 English-language news records, 1,621,432 words, 735 image records and 710 unique downloaded image objects. Raw HTML was kept before extraction, and every record carries a stable identifier, URL, retrieval timestamp and hash.

Validation found no malformed JSON, duplicate document identifiers or missing image files. Five publication dates are missing, eight page dates fall after the 12 July 2026 collection window, and seven pages contain no recoverable article body; these cases are flagged rather than silently corrected. A static, nonexecuting scan found no high confidence malicious payloads, encoded executables, exploitkit patterns or suspicious download URLs. VirusTotal was also used as an external check of the final JSONL file and associated URLs, with no malicious detections returned (VirusTotal 2026). Initial NLP results highlight recurring identity vocabulary and candidate media, lawandorder and culture/identity frames, but these remain retrieval signals rather than validated interpretations.

The text pipeline yielded 66,306 sentences and 116,512 named entities. After removing spaCy stop words, punctuation, numbers, non content parts of speech and audited web boilerplate strings, the final lexical frequencies were rebuilt from 764,245 content tokens comprising 36,930 lemmas. Exploratory keyness against the informative writing frequencies in UCREL BNC List 4.1 identifies recurrent contrasts led by *white*, *black*, *British*, *people* and *Britain* (ranked by log ratio; see below). A searchable dashboard supports filtering, KWIC concordances, collocation exploration and human review. Image analysis is still running, so no visual or multimodal findings are reported yet.

2. How the data was collected

The source was the public Patriotic Alternative website named in the assignment. A preflight check confirmed public access, pagination behaviour, page structure, robots signals and any exclusions. No login, CAPTCHA, hidden endpoint, access-control bypass or rate-limit bypass was used at any point. Because article URLs do not share a reliable `/news/` path, discovery was restricted to headline links across 181 archive pages.

Collection relied on Python 3.13.5, Requests 2.32.5, Beautiful Soup 4.14.3 and `jjsonschema` 4.26.0 (Reitz and Contributors 2026; Richardson 2026; python-jjsonschema contributors 2026). Pages were saved as raw HTML first and reparsed offline, which kept capture and extraction independent. The extractor kept article paragraphs, lists, quotations and subheadings, and dropped reaction widgets, sign in prompts, newsletter blocks and footer material. Hero and inline images, lazy loading fields, the largest `srcset` candidate, explicit captions and `alt` text were stored as separate records.

Three issues came up during development. A fixed minimum-body-length rule was rejecting valid short pages, so it was replaced with structure aware validation. A run that saved HTML but produced no records was recovered once capture and offline reparsing were separated into independent steps. Image coverage was corrected by joining the article and image manifests on `document_id`; content addressed storage then reduced 735 image references to 710 unique objects.

Output	Count	Validation note
Articles	1,801	Valid JSONL lines; no duplicate identifiers
Words	1,621,432	Seven empty-body pages retained for audit
Images	735	710 unique objects; no missing files
Dates	1,796 present	Five missing and eight post-capture dates flagged

The final JSONL file is 33,063,566 bytes, with SHA-256 `083e7fb4ca92631d9ae7ecfc7819586a0410eafb489c52cdab31595af4c08277`. Static inspection confirmed strict UTF-8 encoding, 1,801 valid records, HTTP(S)-only URLs and zero high confidence malicious indicators. The scan raised 61 contextual alerts, all of which turned out to be ordinary prose uses of words such as “shell” and “payload”.

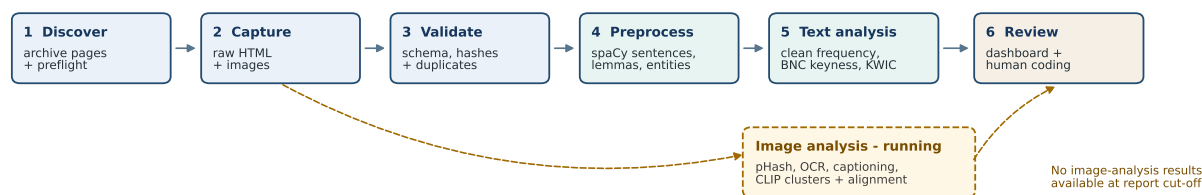


Figure 1: Executed text corpus pipeline and the image analysis is still running

3. Why the data and outputs are organised this way

The article is used as the main unit because details such as the date, title, author, tags and images belong to the full webpage. Paragraphs and sentences are kept under each article so they can still be used for concordance searches and discourse analysis. The repository separates the raw HTML and images from the processed JSONL files, analysis tables and dashboard data. This means the extraction and analysis can be repeated without changing the original source files.

Each article has a stable `document_id`, which links the webpage, metadata, text, sentences, images and analysis results. Each image also has its own `image_id`, allowing several images to be linked to the same article. Missing dates remain null unless they can be recovered from reliable page metadata. Raw text and images remain access controlled until copyright, data-protection and ethics issues are reviewed, while the code, schemas, checksums and aggregate results are made public. The interactive dashboard is available at nouran-khallaf.github.io/pa-news-multimodal-corpus/.

4. What insights the data may support

The spaCy transformer pipeline produced 1,607,528 alphabetic tokens before lexical filtering (Honnibal et al. 2020). Frequency counts were then restricted to lowercase lemmas tagged as nouns, proper nouns, verbs, adjectives or adverbs, with spaCy stop words, single character items and ten documented web boilerplate forms excluded. This left 764,245 content tokens across 36,930 unique lemmas. The most frequent terms are *people* (6,391 occurrences across 1,322 documents), *white* (4,168; 946) and *British* (2,619; 856). Among the cleaned recurrent word sequences are *Patriotic Alternative*, *white people*, *British people*, *mass immigration* and *prime minister*.

Exploratory keyness draws on the informative writing **FrIn** column of the lemmatised UCREL BNC List 4.1 (UCREL 2026). Only lemmas listed in that reference are compared, and missing entries are never treated as zero. Applying minimum thresholds of 20 target occurrences and ten documents left 559 matched terms. Log-likelihood ranks the divergence and log ratio reports the effect size (Rayson and Garside 2000; Hardie 2014). Among the strongest high-frequency contrasts are *white* (log ratio 3.91), *black* (2.60), *British* (1.82), *people* (1.67) and *Britain* (1.53). These figures describe relative frequency, not ideological meaning, and the dashboard links every result back to KWIC lines for human checking.

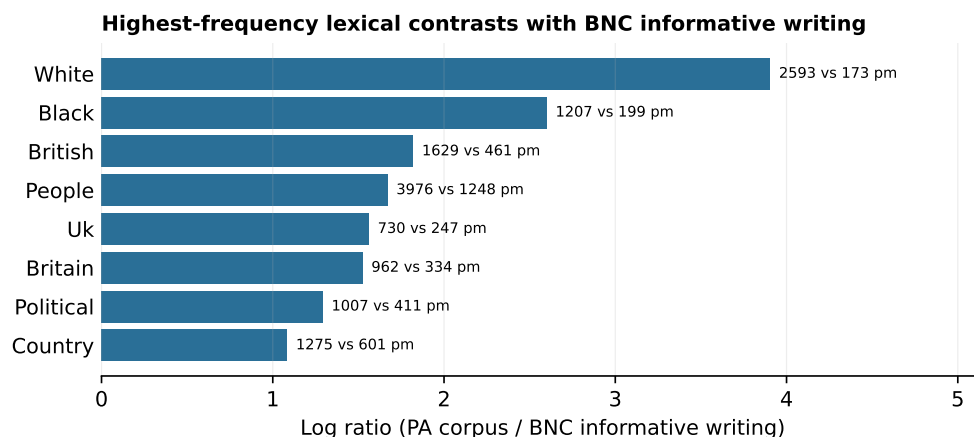


Figure 2: Preliminary overuse relative to BNC informative writing. Bars show log ratio, annotations show target versus BNC frequency per million.

Automated retrieval also produced 78,402 actor mentions, 20,038 frame candidates and 8,939 legitimization candidates, all of which are review queues rather than confirmed findings. Quotation and reporting verb flags are kept to reduce incorrect attribution, and any substantive claim should combine document frequency, collocation, concordance and independent human coding (Baker et al. 2008; van Leeuwen 2007).

The planned image workflow covers perceptual hash reuse, OCR, generated captions, CLIP embeddings and provisional clustering, followed by title/lead/caption alignment and stratified human review (Smith 2007; Radford et al. 2021; Kress and van Leeuwen 2021), face recognition and demographic inference are explicitly excluded. Because GPU image processing was still running when the task deadline arrived, no image-analysis or image/text results could be included in this report.

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